SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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DRAFT - team details, declarations, platform evidence and signature pending

1. TEAM AND PROJECT OVERVIEW

Team Name

Awaiting registered team name

Selected Domain

Domain 2: Orbyte Smart SME Challenge

Selected SDG Track (1–17)

8 - Decent Work and Economic Growth

Project / Solution Name

Orbyte DecisionOS

Name (Leader)

Awaiting team leader confirmation

Email & Phone (Leader)

Awaiting leader email and phone

University Affiliation(s)

Awaiting university affiliation(s)

Number of Team Members

Awaiting registered count (maximum 5)

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2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian micro and small cafes repeatedly check ingredient stock, request quotations and decide when to reorder. The owner often has to reconcile separate records before approving a purchase. Arithmetic mistakes, late ordering and undocumented decisions can disrupt service and add administrative work.

DecisionOS focuses on milk replenishment in a fictional cafe. The reference case has 18 litres in stock, estimated use of 12 litres per day and a three-day minimum coverage threshold. Current coverage is therefore 1.5 days. One supplier offers a lower price but estimates delivery after available stock runs out; another offers earlier delivery at a higher price. Price alone is insufficient for the decision.

The affected users are cafe owners and purchasing managers. The prototype brings stock evidence, delivery assumptions and quotation arithmetic into one transparent approval journey. It aims to reduce repetitive comparison work while preserving owner control and a traceable record.

This is a problem hypothesis supported by the challenge brief and a synthetic scenario, not a completed SME field study. A timed usability comparison is planned. No real customer, employee or company records are used, and no business savings or stockout reductions have been measured.

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2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Orbyte DecisionOS combines a working operational workflow with an optional AI scenario layer. A user initiates an inventory check; our mock Orbyte-style connector retrieves synthetic inventory and quotations. Python rules calculate coverage, cost and delivery risk. An owner selects or rejects an option, records a reason and receives a persistent purchase-order draft and audit trail after approval. No supplier is contacted and no payment is made.

The UI exposes the RM90 premium for earlier estimated milk delivery. Backend role checks, snapshot validation and idempotency protect approval and draft creation.

The optional MiroFish adapter explores a delivery delay or menu-price change through fictional stakeholder perspectives. Its source is pinned and HTTP contracts are tested. A separate environment is installed, but a genuine run needs approved LLM/Zep credentials and budget. The UI shows unavailable status rather than invented results. Agent narratives will not replace arithmetic or become unsupported forecasts.

The approach applies database, API and cloud-deployment concepts relevant to the workshops. Team workshop completion is unconfirmed. Replit configuration is prepared; actual project use and proof remain pending. The priority is a complete operational journey.

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3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: Next.js 16.4.2, React 19.3.1, TypeScript 5.9.3 and Tailwind CSS 4.3.3. Backend: Python 3.12, FastAPI 0.135.1, Pydantic 2.12.5 and SQLAlchemy 2.0.48. SQLite stores business records, decision snapshots, approvals, order drafts, audit events and simulation requests. Money uses integer sen; quantities and rules use Decimal.

Next.js proxies API requests to FastAPI. The mock connector is our own documented interface assumption, not an official Orbyte API. Optional simulation requests are processed by a separate local worker against MiroFish commit 7657031ac01184afe2cb220f5ee3545573b5e843. LLM/Zep services remain disabled pending approval.

Pytest, Vitest, TypeScript and a production build have passed. Playwright tests are implemented but local Chromium startup is blocked by the macOS sandbox. Native-browser checks verified the working journey. See Appendix A for architecture.

Sponsor resources: no actual sponsor API/cloud use is yet evidenced. Replit import configuration is prepared and sign-in is pending. No GPU workload has been executed; RunPod is currently not applicable. Update this statement after genuine platform use.

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3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The design starts with one recurring owner task: deciding which supplier can replenish a critical ingredient in time. A synthetic journey makes the data, trigger and decision explicit before adding broader modules. The interface places quote costs, estimated delivery and missing information beside owner approval controls; roles are clearly labelled as simulated.

The working local prototype has been iterated through automated backend/component checks and direct browser use. Approval, rejection, reload persistence and mobile navigation were checked. A timed study with consenting testers is still planned; no interviews or user validation are claimed.

Phase 2 work will validate the problem with SMEs, measure task completion against a manual baseline, and refine the workflow from observed errors. Real deployment would require authentication, tenant isolation, migrations, backups and a reviewed privacy process. The current synthetic Replit preview is prepared for import, not yet verified there.

The main integration risk is unreliable or costly simulation execution. Deterministic procurement remains usable when MiroFish is offline. A small approved run will be measured before any AI result is demonstrated.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 8, targets 8.2 and 8.3. 8.2 concerns productivity through technological upgrading and innovation. DecisionOS aims to reduce repetitive procurement administration. 8.3 includes productive activities, entrepreneurship and SME growth. Traceable owner-controlled workflows are a proposed contribution, not evidence of policy-level impact. Source: UN DESA, Goal 8, https://sdgs.un.org/goals/goal8 (reviewed 22 Oct 2026).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Estimated planning baseline: 10 minutes to complete a correct manual comparison, approval record and procurement draft for one shortage case. This is a team planning assumption, not an observed measurement or published benchmark. Replace it with the median of actual manual trials before making a measured improvement claim. Reference operational input: 18 / 12 = 1.5 days of stock coverage. This is calculated from synthetic data.

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4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Primary KPI: percentage reduction in median time to complete a correct procurement draft, calculated as (manual median - DecisionOS median) / manual median x 100. Guardrails: arithmetic errors per task; percentage of approved drafts with actor, reason and timestamp; unassisted completion rate. Use identical synthetic tasks, alternate task order and report sample size and help given.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 target: at least 30% lower median task time; at a 10-minute baseline, this means no more than 7 minutes. Guardrails: zero reference-case arithmetic errors, 100% complete approval records and at least 80% unassisted pilot completion. These are proposed targets. No productivity gain, waste reduction, revenue change or stockout probability has been measured.

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5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

Awaiting registered-team review and leader confirmation. No workshop completion, legal declaration or signature has been assumed.

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6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Awaiting each registered member's confirmation. The local DecisionOS repository was created during this build on 21 October 2026; no older application repository was reused in the observed workspace. This does not establish whether team members have other prior-owned work to disclose.

Substantial AI-assisted implementation is declared separately. Third-party frameworks and the separate MiroFish dependency are listed in Section 6.2 and Appendix B. Do not select No until the team has reviewed the prior-work question.

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6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js / React / Tailwind	Application frameworks	MIT
Python / SQLite	Language / database	PSF / public domain
FastAPI / Pydantic / SQLAlchemy	Backend frameworks	MIT
TypeScript / Playwright	Compiler / browser tests	Apache-2.0
MiroFish; other tools	Separate service; see Appendix B	AGPL-3.0; others as listed

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6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

Awaiting registered-team confirmation. Each member must check employment, internship and consulting obligations before the leader selects Yes or No. No agreement or ownership clearance has been inferred from the absence of information.

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7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]

Awaiting registered-team review and leader confirmation. No workshop completion, legal declaration or signature has been assumed.

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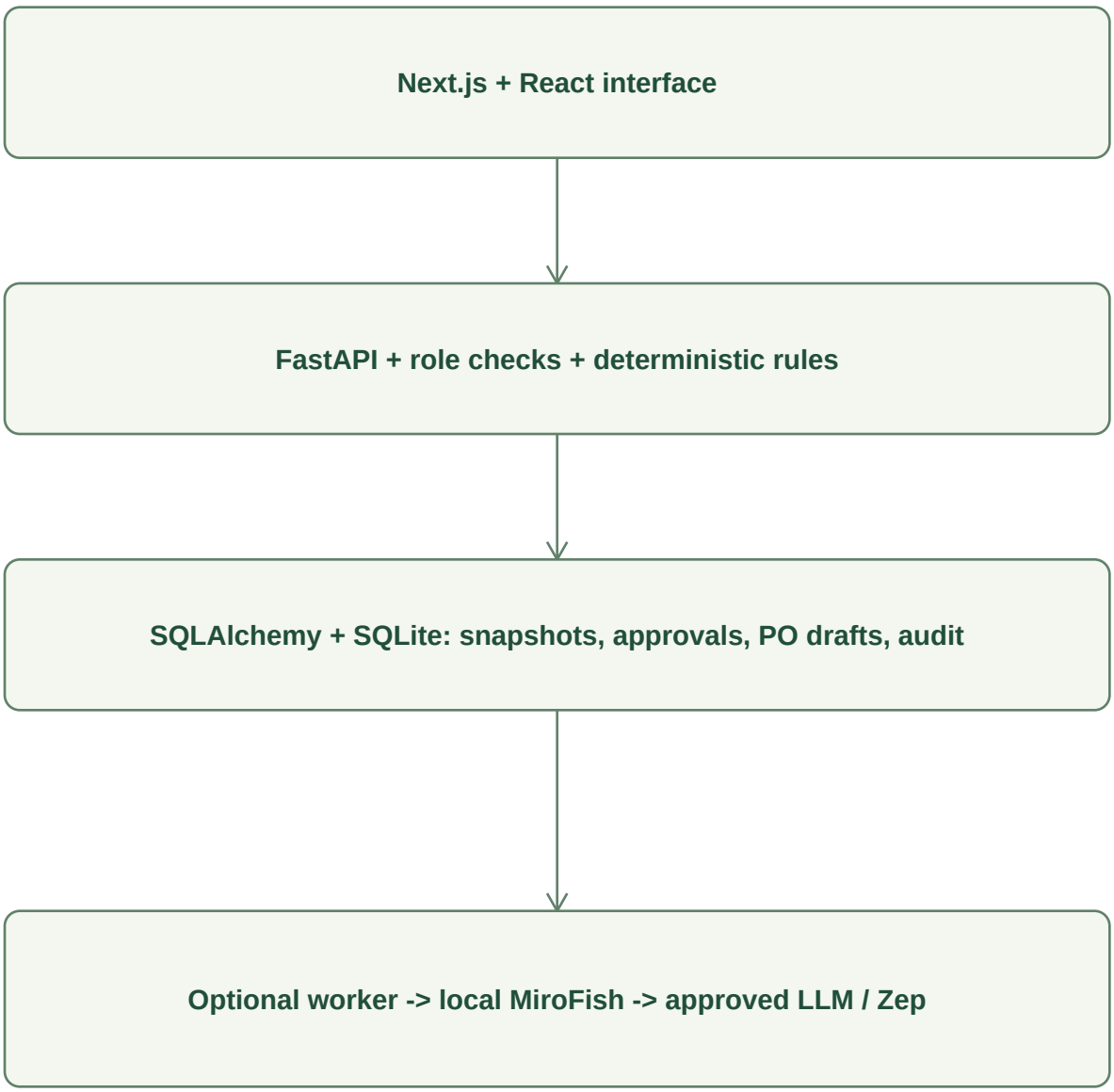
Appendix A: system architecture

Synthetic operational workflow; optional simulation remains disabled.



Our mock Orbyte-style connector supplies business-scoped synthetic inventory and quotes. No official Orbyte API is claimed.

Approval, purchase-order draft and audit records commit together. No real order, payment or inventory receipt occurs.

Replit configuration is prepared; authenticated project use and genuine evidence remain pending.

Appendix B: dependency notices

Python 3.12 / SQLite	PSF / public domain
Next.js 16.4.2 / React 19.3.1 / React DOM 19.3.1	MIT
TypeScript 5.9.3 / Playwright 1.64.1	Apache-2.0
Tailwind CSS / PostCSS plugin 4.3.3	MIT
Lucide React 0.468.0	ISC
FastAPI 0.135.1 / Pydantic 2.12.5 / SQLAlchemy 2.0.48	MIT
Uvicorn 0.41.0 / HTTPX 0.28.1 / python-dotenv 1.2.2	BSD-3-Clause
Pytest 9.0.2 / Vitest 5.0.5 / jsdom 30.1.3	MIT
Testing Library React 16.3.3 / jest-dom 7.0.1	MIT
pytest-asyncio 1.3.0	Apache-2.0
MiroFish 7657031 (separate optional service)	AGPL-3.0

Versions and licences above were checked against installed package metadata and the pinned upstream licence. Full transitive dependency versions are retained in the lockfiles. MiroFish source is not bundled with the Apache-2.0 application. Preserve upstream notices and review AGPL obligations before external deployment.

Development assistance: OpenAI Codex (major use). Whisper base.en via faster-whisper transcribed workshop videos locally for rules review. ReportLab/PyPDF and the Codex presentation runtime generated documents. These research/authoring tools are not application services.